DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST - I

1005237729008

Subject : Graph Theory

Class: B.E (AIML)
Semester: IV
Academic Year: 2024-2025

Max Marks: 30
Duration: 1 hr 30 Minutes
Date : 29-03-2025 (2:30 p.m. to 4:00 p.m.)

Sl. No.	PART A Answer All Questions (5 x 2 = 10 Marks)	Marks	BT	CO
1.	Is there a simple graph with the degree sequence (1,1,3,3,3,4,6,7).	2	2	1
2.	Determine all non-isomorphic simple non-directed graphs of order 3 (Hint: |V| is order of graph).	2	3	1
3.	An edge 'e' in a connected graph G is cut edge if G-e is not connected. Prove that a connected graph is a tree if only iff each edge is cut-edge.	2	3	2
4.	If a graph G is not connected. Prove that its complement G' (G Complement) is connected . Is the converse is true.	2	3	2
5.	i) Define Bipartite. ii) Specify whether the following graph are bipartite are not. Star graph, W_n , K_n, $K_{2,3}$	2	1 1	1 2
	PART B **Answer Any Two Questions (2 x 10 = 20 M)**			
6.	Write BFS algorithm and explain with the help of an example	10	2	2
7.	a) State and prove the sum of degree theorem for non-directed graph.	5	2	1
	b) Prove that in any non-directed graph there is an even number of vertices of odd degree.	5	2	1
8.	Draw the conditions under which a graph G is isomorphic to graph H	10	3	1

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